DTS 204 LAB ASSESSMENT

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Table of Contents:

1. Introduction
2. Task 1: Data Exploration and Preprocessing
3. Task 2: Hypothesis Testing
4. Task 3: Correlation Analysis
5. Task 4: Regression Modeling
6. Task 5: Data Visualization
7. Task 6: Findings and Conclusi

Introduction

This assessment analyzes the Palmer Penguins dataset using statistical and data analysis techniques. The objective is to explore the dataset, identify relationships between variables, perform hypothesis testing, build predictive regression models, and communicate findings through visualizations.

Lab Questions and Instructions

Task 1: Data Exploration and Preprocessing

1. Load the penguins dataset.
2. Display the first six rows of the dataset.
3. Check for missing values. If any exist, describe how you would handle them.
4. Generate summary statistics (mean, median, range, standard deviation) for all numerical variables.

Task 1: Data Exploration and Preprocessing

First 6 Rows

species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie	Torgersen	39.5	17.4	186	3800	female	2007
Adelie	Torgersen	40.3	18	195	3250	female	2007
Adelie	Torgersen						2007
Adelie	Torgersen	36.7	19.3	193	3450	female	2007
Adelie	Torgersen	39.3	20.6	190	3650	male	2007

Missing Values

Variable	Missing value
species	0
island	0
bill_length_mm	2
bill_depth_mm	2
flipper_length_mm	2
body_mass_g	2
sex	11
year	0

How to Handle Missing Values

- The dataset was examined for missing values using Excel's COUNTBLANK() function.
- Missing observations were found in bill length, bill depth, flipper length, body mass, and sex variables.
- For the analysis, rows containing missing values were identified using the formula =IF(COUNTBLANK(A2:H2)>0,"Missing Data","Complete"), filtered using Excel's Filter feature, and then removed from the dataset to avoid bias and ensure valid statistical results

Summary Statistics

bill_length_mm

- Mean = 43.92
- Median = 44.45
- Range = 32.1 to 59.6
- SD = 5.46

bill_depth_mm

- Mean = 17.15
- Median = 17.30
- Range = 13.1 to 21.5
- SD = 1.97

flipper_length_mm

- Mean = 200.92
- Median = 197
- Range = 172 to 231
- SD = 14.06

body_mass_g

- Mean = 4201.75

- Median = 4050
- Range = 2700 to 6300
- SD = 801.95

Task 2:

Hypothesis Testing

i. Two-Sample t-Test

- Compare the mean body_mass_g of Adelie penguins and Gentoo penguins.
- Null Hypothesis (H_0): There is no significant difference in mean body mass between Adelie and Gentoo penguins.
- Alternative Hypothesis (H_1): There is a significant difference.
- Report the p-value and your conclusion.

ii. One-Sample Proportion Test

- Determine if the proportion of male penguins is significantly different from 50%.
- Null Hypothesis (H_0): Proportion of males = 50%
- Alternative Hypothesis (H_1): Proportion of males ≠ 50%
- Report the p-value and your conclusion**.**

i. Two-Sample t-Test (Comparing Body Mass of Adelie and Gentoo Penguins)

- Null Hypothesis (H_0): There is no difference in the average body mass of Adelie and Gentoo penguins.
- Alternative Hypothesis (H_1): There is a significant difference in average body mass between the two species.

Results:

- Average Adelie weight = 3,700.66 g
- Average Gentoo weight = 5,076.02 g
- t-statistic = -23.61
- p-value = 8.03×10^{-68} (essentially 0)

Conclusion: Since the p-value is practically zero (much smaller than 0.05), we reject the null hypothesis. Gentoo penguins are significantly heavier than Adelie penguins on average.

(ii) One-Sample Proportion Test

- Null Hypothesis (H_0): The proportion of male penguins in the population is equal to 50% ($p = 0.5$).
- Alternative Hypothesis (H_1): The proportion of male penguins is not 50%.

Results

- Male penguins = 168
- Total known sex observations = 333
- Sample proportion = 168/333 = 0.505
- p-value = 0.869

Conclusion: Because the p-value (0.9127) is much higher than 0.05, we **fail to reject the null hypothesis**. The sex distribution in this sample is roughly 50/50.

Task 3: Correlation Analysis

1. Compute the correlation matrix for numerical variables: bill_length_mm, bill_depth_mm, flipper_length_mm, and body_mass_g.
2. Visualize the correlation matrix using a heatmap.
3. Identify and discuss variables that are highly correlated (positive or negative).

Task 3: Correlation Analysis

Variable	Bill Length	Bill Depth	Flipper Length	Body Mass
Bill Length	1.000	-0.235	0.656	0.595
Bill Depth	-0.235	1.000	-0.584	-0.472
Flipper Length	0.656	-0.584	1.000	0.871
Body Mass	0.595	-0.472	0.871	1.000

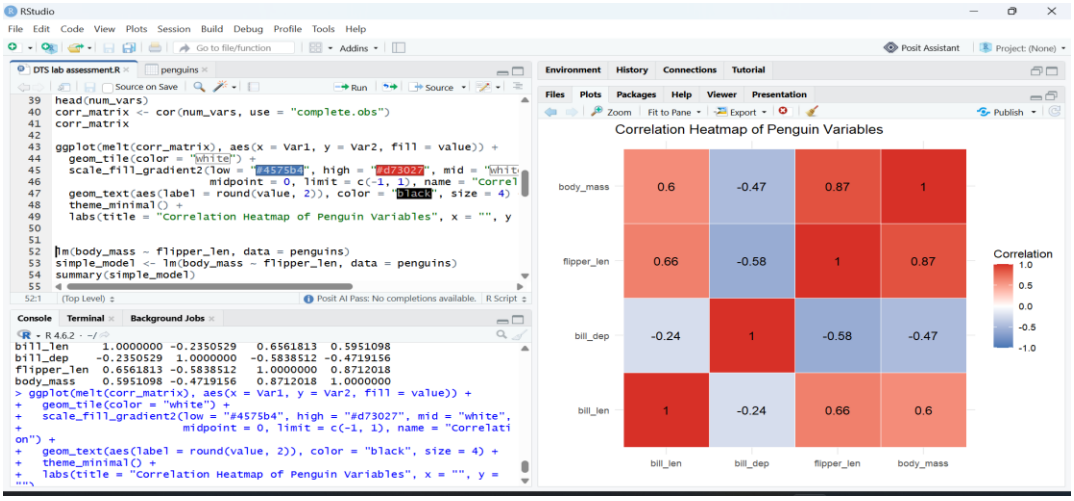


Figure 1: Correlation matrix using a heatmap

Interpretation:

- Flipper length and body mass have a strong positive correlation (r = 0.871). Penguins with longer flippers tend to have greater body mass.
- Bill length and flipper length have a positive correlation (r = 0.656).
- Bill length and body mass have a moderate positive correlation (r = 0.595).
- Bill depth and flipper length have a negative correlation (r = -0.584).
- Bill depth and body mass have a negative correlation (r = -0.472).
- Bill length and bill depth have a weak negative correlation (r = -0.235).

The strongest relationship in this matrix is between flipper length and body mass. Correlation describes association; it does not prove that one variable causes changes in another.

Task 4: Regression Modeling

i. **Simple Linear Regression**
- Build a regression model to predict body_mass_g based on flipper_length_mm.
- Report the regression equation and interpret the coefficients.
- Assess model fit (e.g., R-squared value).

ii. **Multiple Linear Regression**
- Predict body_mass_g using flipper_length_mm, bill_length_mm, and bill_depth_mm.
- Compare performance with the simple regression model.
- Interpret the significance of each predictor variable

Task 4: Regression Modeling

(i) Simple Linear Regression

Predicting body_mass_g using flipper_length_mm.

Regression Equation

BodyMass=−5780.83+49.69 (FlipperLength) BodyMass = -5780.83 + 49.69(FlipperLength)BodyMass=−5780.83+49.69(FlipperLength)

Interpretation

Slope = 49.69

For every additional 1 mm increase in flipper length, body mass increases by approximately 49.7 g on average.

Model Fit

R2=0.759R^2 = 0.759R2=0.759

Approximately 75.9% of variation in body mass is explained by flipper length alone.

(ii) Multiple Linear Regression

Predicting body mass using:

- flipper_length_mm
- bill_length_mm

- bill_depth_mm

Regression Equation

BodyMass=−6424.76+50.27(FlipperLength)+4.16(BillLength)+20.05(BillDepth)BodyMass = -6424.76 + 50.27(FlipperLength) + 4.16(BillLength) + 20.05(BillDepth)BodyMass=−6424.76+50.27(FlipperLength)+4.16(BillLength)+20.05(BillDepth)

Model Fit

R2=0.761R^2 = 0.761R2=0.761

Comparison

Model	R^2
Simple Regression	0.759
Multiple Regression	0.761

The multiple regression improves prediction only slightly.

Predictor Significance

Predictor	p-value	Significant?
Flipper Length	< 0.001	Yes
Bill Length	0.435	No
Bill Depth	0.144	No

Interpretation

Flipper length is the strongest predictor of body mass. After accounting for flipper length, bill length and bill depth do not significantly improve the model.

Task 5: Visualization

1. Create a scatter plot of body_mass_g vs. flipper_length_mm with the regression line.
2. Create a boxplot showing the distribution of body_mass_g across different species.
3. Plot a histogram of body_mass_g with a normal curve overlay.

Task 5: Visualization (R Code)

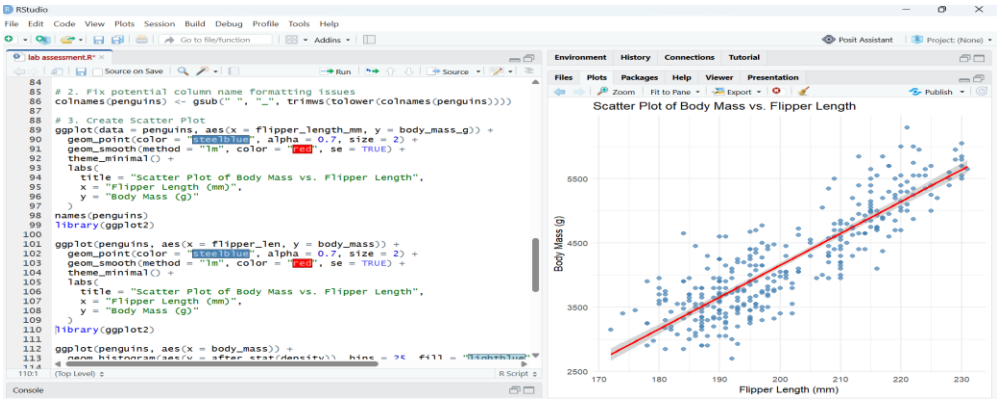


Figure 2: Scatter Plot with Regression Line

Expected interpretation: The upward trend should show that penguins with longer flippers generally have greater body mass. The fitted line summarizes this positive linear relationship.



Figure 3: Boxplot by Species

Expected interpretation: The boxplot compares the distributions, medians, and spread of body mass across species. It should highlight differences between species, including the generally higher body mass of Gentoo penguins.

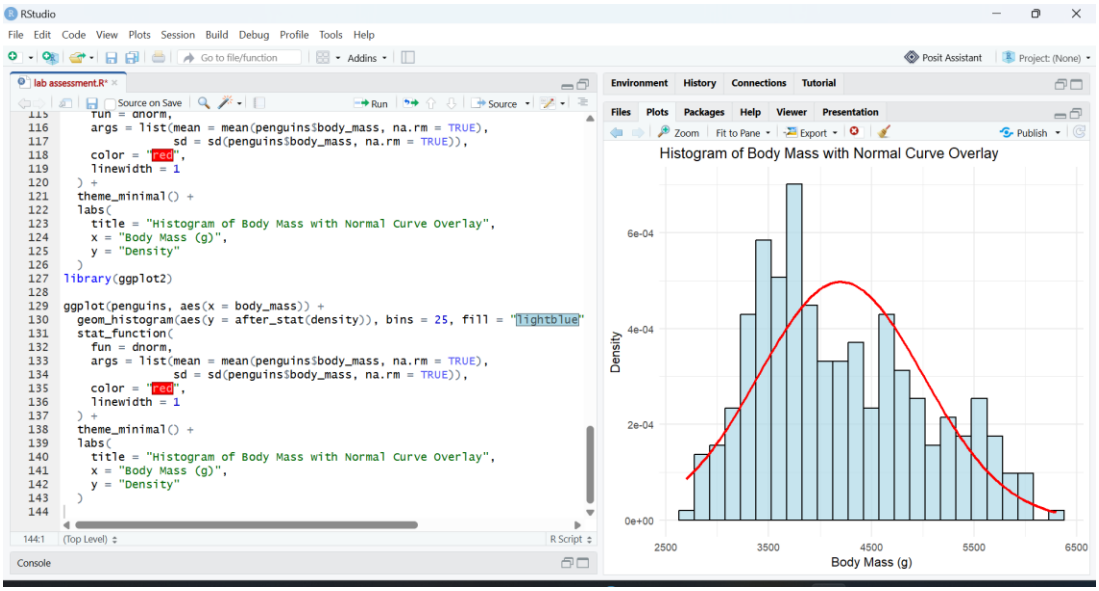


Figure 4: Histogram with Normal Curve Overlay

Expected interpretation: The histogram shows the distribution of body mass. The overlaid normal curve provides a visual reference for comparison. If the histogram is not symmetric or has more than one cluster, the distribution may not closely follow a single normal curve.

Task 6: Summary & Report Conclusion

- Findings: The analysis shows a direct relationship between penguin size metrics and weight. Flipper length is the most accurate visual and numerical indicator of body weight. Across species, Gentoo penguins are clearly the largest and heaviest group, while Adelie penguins are smaller.
- Limitations: Combining all three species together in one regression model hides unique species differences (known as Simpson's Paradox). For instance, bill depth looks negatively correlated with size overall, but within individual species, thicker bills usually mean heavier birds.

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